Flynn S. Harms

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EDUCATION

California State University, Long Beach | Expected Graduation: **May 2027**
Bachelor of Science, Electrical Engineering.
GPA: 3.96/4.0 | Honors Program
Activities and Societies: IEEE and Tau Beta Pi Member
Presidents and Dean's List 2025-2026

SKILLS

- Programming and Modeling: MATLAB, Simulink, C++, Python, VHDL/FPGA Design, Assembly, and AutoCAD.
- Technical Skills: Analog/Digital Circuit design and analysis, Altium, digital signal processing, and machine learning.
- Lab Equipment: Oscilloscopes, Multimeter, Function generator, LabVIEW, MyDAQ, etc.

Relevant Courses – Signals and Systems, Digital Signal Processing, Communication Systems, Control Systems, and Digital Logic Design, and DSP with Machine Learning.

ENGINEERING EXPERIENCE

Southern California Edison – *Protection Standards Engineer Intern*, Pomona, CA | May 2026 – Present

- Develop and revise protection and control standards supporting safe and reliable operation of electric-grid systems.
- Perform laboratory testing to verify protection schemes, relay behavior, and high-impedance fault.
- Configure and evaluate protective-relay settings using engineering calculations, test data, and applicable technical standards.
- Updated 35+ protection standards for a panel retrofit, programming relays, developing logic diagrams, and verifying outputs through lab testing.
- Collaborate with engineers to define technical requirements, resolve design issues, and maintain system documentation.

CSULB Neural Network and Multidisciplinary DSP Laboratory – *Undergraduate Research Assistant* | May 2026 – Present

- Develop a machine-learning and digital-signal-processing workflow to detect high-impedance faults in distribution-level circuits using Southern California Edison data.
- Process, clean, and analyze electrical-system data to identify fault signatures and evaluate detection performance.
- Implement and compare detection methods in MATLAB and Python under normal and abnormal operating scenarios.
- Define test cases and verification criteria to evaluate model accuracy, robustness, and false-detection behavior.
- Collaborate with faculty and utility engineers to define technical objectives, document methods, and interpret results.

CSULB College Of Engineering – Engineering Tutor | January 2026 – August 2026

- Tutor students in EE 310 Signals and Systems, EE 330 Analog Circuits, EE 201 Digital Logic, and PHYS 254 Modern Physics.
- Explain technical concepts and guide students through circuit analysis, Fourier analysis, Boolean logic, finite-state machines, and physics problem solving.
- Prepare students for exams and assignments through individualized instructions, worked examples and targeted review.

PROJECTS

FPGA-Based Digital Combination Lock | VHDL – April 2026

- Designed and implemented a synchronous digital combination lock in VHDL using a finite-state-machine architecture.
- Defined functional behavior and interfaces for password entry, verification, reset, lockout, and status outputs.
- Developed a VHDL testbench and verified normal operation, incorrect entries, reset behavior, and edge cases through simulation in Xilinx Vivado.
- Synthesized and deployed the design using Xilinx Vivado on an Artix-7 FPGA development board, integrating switches, pushbuttons, LEDs, and seven-segment displays.

AC/DC Power Conversion and Battery-Backup System — Senior Design Capstone – In Progress

- Developing a self-contained power system that converts 120 V, 60 Hz AC input into regulated DC and low-power AC outputs.
- Translating system specifications into requirements and interfaces for rectification, EMI filtering, DC/DC conversion, battery charging, PWM inversion, sensing, and protection.
- Constructed and verified a full-bridge rectifier through simulation and evaluated components based on voltage, current, ripple, and load requirements.
- Conducting topology trade studies to balance efficiency, voltage regulation, safety, component ratings, and system reliability.